Raslen Ziadi

Electrical Engineer | Embedded Systems
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Professional Profile

Electrical engineer specializing in embedded software, microcontroller development, test automation, and AI-assisted validation. Experienced in C, C++, Python, Robot Framework, embedded Linux, CI/CD, and intelligent diagnostic solutions for reliable software systems.

Technical Skills

Programming & scripting: C, C++, Python, Java, Shell
Embedded software: STM32 (HAL/LL), ARM Cortex-M, FreeRTOS, ZephyrRTOS
Test automation: Robot Framework, Selenium, PyAutoGUI, regression testing, log analysis, automated reporting
AI & Machine Learning: LLM API integration, AI-assisted diagnostics, incident classification, Machine Learning
DevOps & version control: Git, GitLab, GitHub, Jenkins, CI/CD, continuous testing
Embedded Linux & IoT: Yocto, QEMU, MQTT, Linux scripting
Protocols & tools: UART, SPI, I2C, CAN, STM32CubeIDE, PyCharm, VS Code, Qt

Professional Experience

STMicroelectronics – Final-Year Internship, Feb. 2026 – Aug. 2026

- Designed a modular Python/Robot Framework setup for automated STM32 Secure Boot validation.
- Automated compilation, provisioning, hardware reset, UART capture, regression testing, and reporting.
- Reduced validation time by up to 48% (approximately 15 to 8 minutes), enabling continuous and reproducible testing.
- Implemented AI-assisted fault diagnosis using Robot Framework/UART logs, JSON incidents, local rules, and an LLM API.

Technologies: Test Automation, Robot Framework, LLM integration, validation, reporting, Python, STM32, UART, JSON.

LAS2E – Laboratory for Automation, Electrical Systems and Environments, ENIM – Engineering Internship, Jul. 2025 – Aug. 2025

- Developed a microcontroller-based prototype to monitor a motor's energy consumption.
- Deployed an embedded Linux monitoring solution using Yocto on Raspberry Pi 4.
- Integrated sensors and MQTT communication for an embedded IoT monitoring system.
- **Technologies:** STM32, sensors, Yocto, Raspberry Pi 4, MQTT, IoT.

STEG Menzel Bouzelfa – Introductory Internship, Jul. 2024 – Aug. 2024

- Inspected industrial electrical systems in accordance with safety procedures.
- Participated in observation and maintenance activities in an industrial environment.
- **Areas:** industrial electrical systems, maintenance, safety.

Projects

Yocto-Based Multi-Sensor Embedded IoT System – Real-time monitoring, data acquisition, MQTT, alarms, and logging.
Technologies: Yocto Linux, IoT, MQTT, Python, QEMU

Driver Assistance System – STM32-based module integrating GPS and LiDAR for navigation and obstacle detection.
Technologies: STM32, LiDAR, GPS, graphical interface

Autonomous Robot (Eurobot) – STM32- and ROS2-based autonomous robot capable of planning and executing multiple tasks simultaneously.
Technologies: STM32, FreeRTOS, CAN, SPI

Zephyr RTOS-Based Embedded Monitoring and Alarm Management System – Developed a modular embedded monitoring system with concurrent tasks, simulated sensor data, threshold-based alarms with hysteresis, and Python visualization.
Technologies: C, Zephyr RTOS, STM32, multithreading, alarm management, Python, embedded systems

STM32H7-Based Electrical Control Cabinet – Designed a control cabinet to operate motors and actuators in an automated environment.
Technologies: STM32H7, TouchGFX, embedded HMI, industrial control

Education

ENIM – National Engineering School of Monastir, 2023 – 2026
Electrical Engineering, Monastir, Tunisia

IPEIEM – Preparatory Institute for Engineering Studies of El Manar, 2021 – 2023
Physics-Chemistry, Tunis, Tunisia

Certifications

nRF Connect SDK Intermediate – Nordic Semiconductor Academy, October 8, 2026
Learning Git and GitHub – LinkedIn Learning, July 2026
Learning Jenkins – LinkedIn Learning, May 2026
Introduction to Linux (LFS101) – The Linux Foundation, November 2025
Open Source RT-Thread RTOS on RISC-V (LFD123) – The Linux Foundation, November 2025
Foundational C# with Microsoft – freeCodeCamp / Microsoft, June 2024

Extracurricular Activities

ING.ENIM Robotics Club – *President, Monastir, Tunisia*, June 2024 – June 2025
Organized events and provided technical guidance in embedded systems and robotics.

Languages

French: Fluent | English: Fluent | Arabic: Native